Cloud Computing & Distributed Systems Architecture

Department: Information Technology | Semester: 6 | Code: IT603
CampusVerse Peer-Reviewed Academic Document

1. Service Models (IaaS, PaaS, SaaS)

IaaS provides virtualized hardware (AWS EC2), PaaS provides runtime platforms (Google App Engine), and SaaS provides ready applications (Google Workspace, Office 365).

2. Microservices & Container Orchestration

Docker containers decouple application code from host dependencies. Kubernetes orchestrates deployments, scaling, and rolling updates across container clusters.

3. CAP Theorem & Distributed Data Stores

A distributed data store can guarantee at most two of three properties: Consistency, Availability, and Partition Tolerance.